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Bihar Engineering University, Patna

End Semester Examination - 2022

Subject: Mathematics-III (Differential Calculus)

Code: 100311

Time: 03 Hours

Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (Any seven question only): **[14]**

(a) The value of $\lim_{x\to 0}(\sin x/x)^{1/x}$ is

(i) 0 (iii) e
(ii) 1 (iv) $1/e$

(b) The value of the integral $\int_c\{yz\,dx + (xz+1)dy + xy\,dz\}$ where C is any path from (1, 0, 0) to (2, 1, 4) is

(i) 6 (iii) 8
(ii) 7 (iv) 9

(c) The maximum value of $\sin x + \cos x$ is

(i) 1 (iii) $\sqrt{(2)}$
(ii) 2 (iv) 0

(d) The value of $\nabla^2[(1-x)(1-2x)]$ is equal to

(i) 2 (iii) 4
(ii) 3 (iv) 6

(e) The degree of the differential equation $y(dx/dy)-(dx/dy)^2-\sin y(dx/dy)^3-\cos x = 0$ is

(i) 0 (iii) 2
(ii) 1 (iv) Cannot be determined

(f) If $f = \tan^{-1}(y/x)$, then div (grad f) is equal to

(i) 1 (iii) 0
(ii) -1 (iv) 2

(g) If P_n is the Legendre polynomial of first kind, then the value of $\int_{-1}^{1} x\,P_nP_n'\,dx$ is

(i) [2] / [(2n+1)] (iii) [2] / [(2n+3)]
(ii) [2n] / [(2n+1)] (iv) [2n] / [(2n+3)]

(h) If J_n is the Bessel's function of first kind, then the value of $J_{-1/2}$ is

(i) $\sqrt{(2/\pi x)}(\cos x/x-\sin x)$ (iii) $\sqrt{(2/\pi x)}\sin x$
(ii) $\sqrt{(2/\pi x)}(\sin x/x-\cos x)$ (iv) $\sqrt{(2/\pi x)}\cos x$

(i) The solution of $p\tan x + q\tan y = \tan z$ is

(i) $\sin x/\sin y = \varphi(\sin y/\sin z)$ (iii) $\sin x/\sin y = \varphi(\sin y,\sin z)$
(ii) $\sin x\cdot\sin y = \varphi(\sin y/\sin z)$ (iv) $\sin x/\sin y = \varphi(\sin y\cdot\sin z)$

(j) The vector $v = e^x\sin y\,i + e^x\cos y\,j$ is

(i) Solenoidal (iii) rotational
(ii) irrational (iv) cannot be found

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Q.2 Solve both questions : **[14]**

(a) Form the partial differential equation $(x-a)^2+ (y-b)^2+ z^2= 1$.

(b) Solve $xp + yq = 3z$.

Q.3 Solve both questions : **[14]**

(a) Find the directional derivative of $\varphi= z^2yz + 4xz^2$ at the point (1, -2, 1) in the direction of the vector $2i-j- 2k$.

(b) Find a unit vector normal to the surface $x^3+ y^3+ 3xyz = 3$ at the point (1, 2, -1).

Q.4 Solve both questions : **[14]**

(a) Solve partial differential equation $y^2z/xp + xzq = y^2$.

(b) Show that the function $f(x,y) =[xy] / [\sqrt{(x^2+y^2)}]$, & $(x,y)\neq(0,0)$
0, & $(x,y) = (0,0)$ is continuous at origin.

Q.5 Solve both questions : **[14]**

(a) If $f = (x^2+y^2+z^2)^{-n}$ then find div grad f and determine n, if div grad $f = 0$.

(b) Verify Green's theorem for $\int_C\{(xy+y^2)dx + x^2dy\}$ Where C is bounded by $y=x$, $y=x^2$.

Q.6 Solve both questions : **[14]**

(a) Evaluate the integral by changing the order of integration $\int_0^{\infty}\int_0^x x\, e^{-x^2/y}dy\, dx$.

(b) Solve the differential equation $(x^2+y^2+x)dx - (2x^2+2y^2-y)dy = 0$.

Q.7 Solve this question : **[14]**

Verify the stokes' theorem for $A= (y-z+2)i+ (yz+4)j- xzk$ Where S is the surface of the cube $x=0$, $y=0$, $z=0$, $x=2$, $y=2$ and $z=2$ above the xy-plane.

Q.8 Solve both questions : **[14]**

(a) Prove that $2nJ_n(x) = x(J_{n-1}(x) + J_{n+1}(x))$.

(b) Prove that $\sum_{n=0}^{\infty}[x^{n+1}] / [n+1]P_n(1) =1/2\log([1+x] / [1-x])$.

Q.9 Solve both questions : **[14]**

(a) Using Green's theorem, evaluate $\int_C[(y-\sin x)dx +\cos x\, dy]$ where C is the plane triangle enclosed by the lines $y=0$, $x=\pi/2$ and $y=2x/\pi$.

(b) Prove that $\text{div}(r^nr) = (n+3)r^n$. Hence show that div (r/r^3) is solenoidal.

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